Developing and Operationalising Natural Language Processing solutions for Child Safeguarding: A Critical Review of Technical Possibilities and Organisational Realities in UK Policing – TITLE TO BE CONFIRMED!!!

Theoretical Perspectives of Advanced Professional Practice - MOD006047

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Abstract

Purpose: Lorem impsum...

Method: Lorem impsum...

Findings: Lorem impsum...

Conclusion: Lorem impsum...

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Acronyms

ACL Association for Computational Linguistics. 11

ACM Association for Computing Machinery. 11

AI Artificial Intelligence. 12

CEOP Child Exploitation and Online Protection Centre. 16

CSA Child Sexual Abuse. 6–8, 16

EU European Union. 12

ICO Information Commissioner's Office. 16

IEEE Institute of Electrical and Electronics Engineers. 11

NIST National Institute of Standards and Technology. 16

NLP Natural Language Processing. 6–8, 10, 11, 16

NPCC National Police Chiefs' Council. 16

PKMS Personal Knowledge Management System. 9

SSRN Social Science Research Network. 12

ZK Zettelkasten. 9

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Chapter 1

Introduction

1.1 Research context

This paper defines the research context before selecting the most appropriate methodology and strategy for the literature review which follows.

Empirical observations, supported by academic and grey literature, (Finkelhor et al., 2024; National Police Chiefs' Council, 2025) identify that reported instances of Child Sexual Abuse (CSA) are increasing, with resultant demand on policing outstripping proactive preventative capabilities.

The gap between demand and capability in both CSA and other operational areas of policing is acknowledged across the sector, generating political and organisational appetite for technological solutions, (National Police Chiefs' Council, 2026). Despite this appetite, evidence from both professional and academic sources identifes that policing is structurally complex and requires reform to better adopt technology, (Cooper, 2024), with cultural barriers further diminishing the successful use of technological solutions, (Thompson and Manning, 2021; Kassem and Erken, 2025).

These challenges are significant, yet they do not negate the potential of emerging technologies to address the identified capability gap. Advances in Natural Language Processing (NLP) technologies present an opportunity to deliver preventative safeguarding in the CSA context, acknowledging that the gap between technical possibility and operational deployment is not primarily technical but organisational.

This paper explores how to bridge this gap, drawing on four literature tracks to develop a conceptual framework capable of addressing the proposed research challenge: *How can natural language processing methods be developed and operationalised to identify precursors of contact sex offending against children, to enhance proactive safeguarding opportunities within the existing operational constraints of policing?*

This challenge consists of four thematic research areas:

1. What linguistic precursors of contact sex offending exist in offender-victim conversations?
2. Which NLP methods can be most effectively realised within policing infrastructure?
3. What implementation barriers exist within policing that may hinder NLP adoption?

4. What strategies can be developed to overcome those barriers?

1.2 Purpose and scope of the paper

This research is inherently interdisciplinary, positioned at the intersection of technical and organisational domains. Accordingly, a literature review will require a multi-track approach, with each track addressing a different aspect of the research challenge. The four tracks are as follows:

1. The first track explores the landscape of CSA and the linguistic characteristics which precede contact offending, establishing the empirical foundation for a technical solution.
2. The second track reviews the technical literature on NLP applications in safeguarding and criminal justice contexts, focusing on grooming language detection and the computational methods available for identifying linguistic precursors of contact offending.
3. The third track examines the organisational dynamics of technology adoption in policing, drawing on both academic literature and empirical evidence of structural and cultural barriers to technical innovation.
4. The fourth track addresses the ethical and governance boundaries within which any NLP solution must operate, encompassing algorithmic bias, data protection, proportionality, and the ethical context surrounding preventative safeguarding.

Each track is explored in depth, and their insights are then synthesised to identify where technical possibilities meet organisational realities. This structure enables a comprehensive understanding of both the capabilities and constraints shaping both the research problem and the practice improvement opportunity and grounding any solution in organisational and ethical realities

Chapter 2

Literature Search Strategy and Methodology

2.1 Review Methodology and Rationale

This literature review adopts a systematic integrative approach. Snyder (2019) documents a number of review types, noting that an integrative review seeks to combine multiple perspectives, synthesise those views, and to develop a new framework or perspective.

The interdisciplinary nature of this research, noted in section 1.2, necessitates a review methodology that can accommodate multiple perspectives and evidence types. An integrative design as defined by Torraco (2016) and Torraco (2005) allows for a review and synthesis across domains, building a conceptual framework which seeks to unify the disparate domains into a framework for operationalising and implementing an NLP solution into a safeguarding CSA context.

A systematic approach is layered into this integrative approach to allow for a targeted and transparent search and selection process in reviewing the evidence, ensuring that the review is both comprehensive and replicable. This systematic layer is particularly important given the breadth of this review, which spans technical, organisational and ethical domains, each with their own bodies of literature and evidence types. Whittemore and Knafl (2005) identifies that by adding a systematic approach to an integrative review, findings and subsequent synthesis across domains can lead to the delivery of evidence based practice improvements.

From an academic perspective, a doctoral literature review needs to be defensible and reproducible. Whilst the integrative approach allows for synthesis and framework development, a layered systematic approach ensures reproducibility whilst adding a counter balance to my own personal biases and my tendency to approach problems with a solutionist mindset.

2.2 Tools and software

The multifaceted nature of this review demands a robust literature and note management strategy supported by appropriate tools. Basu (2020) describes the Zettelkasten (ZK) method which can be used to manage literature sources and create links between them, supporting the synthesis process. This is furthered by Reck (2023) who describes the ZK method as a process of creating notes on individual literature sources, applying tags to those notes and then using the tagging system to create links between the notes. Both Basu (2020) and Reck (2023) describe the ZK method as the process of building a Personal Knowledge Management System (PKMS). Cheong and Tsui (2011) describes a PKMS as a structure which allows for the development of "*personal information management, personal knowledge internalization, personal wisdom creation, and interpersonal knowledge transferring*"

Reck (2023) considers the use of various software tools to support the ZK method. Following a period of experimentation with various applications, a software stack to support the ZK method for this review was selected consisting of Zotero for literature management and bibliography generation, Obsidian for note taking and synthesis and LaTeX for writing and formatting the review.

Zotero was selected for its robust literature management capabilities, including the ability to capture metadata, PDFs and notes on literature sources, and to auto generate and update a master bibliography file. Obsidian was selected for its flexibility in note taking and synthesis, allowing for the creation of a network of linked notes which can be easily navigated and searched. LaTeX was selected for its professional formatting capabilities, particularly for handling citations, figures and tables, and for its ability to draw directly from Zotero's auto-generated bibliography, thus ensuring consistency and ease of referencing.

All content is then automatically synced to a cloud store and defined content saved a second time to a GitHub repository. My use of the ZK method and the associated software stack is illustrated in figure 2.1.

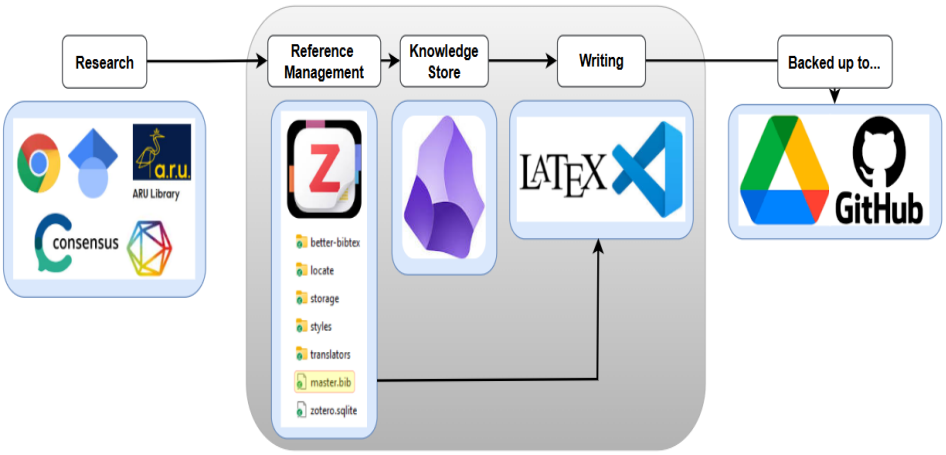

Figure 2.1: Zettelkasten method application for literature and note management

2.3 Search approach and databases

Section 1.2 identifies four distinct research tracks. Each of these tracks has a unique focus and therefore requires a dedicated search strategy to maximise the relevance and quality of included literature. The search strategy presented in table 2.1 defines the four core research tracks, the question to be answered for each track, the selected databases for each track, and the rationale for selecting those databases. The search strategy is designed to ensure that the literature review is comprehensive, targeted and relevant to the research challenge and that it can be replicated by other researchers. The search strategy is also designed to ensure that the literature review is manageable, given the breadth of the research challenge and the interdisciplinary nature of the review.

2.4 Keywords and search terms

Key words and search terms are critical to the success of a literature review. Chigbu et al. (2023) provides holistic guidance on keyword selection and submission to selected databases. The keywords and search terms utilised for each of the noted research tracks are defined in table 2.2. These align to the search strategy outlined in table 2.1. The keywords and search terms are designed to capture the key concepts and themes relevant to each track, ensuring that the literature retrieved is relevant to the research challenge. Boolean operators (AND, OR, NOT) are used to combine keywords and search terms in a way that maximises the relevance of retrieved literature. The use of Boolean operators is critical to refining search results and ensuring that the literature review is focused on the most relevant sources. The specific Boolean operators used for each track are also defined in table 2.2.

2.5 Inclusion and exclusion criteria

Tables 2.3 and 2.4 define the inclusion and exclusion criteria for each of the four research tracks. These criteria are designed to ensure that the literature included in the review is relevant, high quality, and aligned with the research challenge. The notable date of 2018 is presented as this is largely considered to be the breakout year for the devlopment of NLP transformer based models, (Khanna, 2024), an architecture which will be relevant to this research.

Table 2.1: Literature Review Search Matrix

Analytical Question	Databases	Database Justification
Track 1 — Criminology		
What linguistic and behavioural precursors of contact sex offending exist in offender–victim conversations, and how do grooming typologies characterise the progression toward contact abuse?	• Scopus • PsycINFO • Web of Science • Criminal Justice Abstracts • ProQuest • Google Scholar	• **Scopus** — Broad peer reviewed coverage across policing and criminology topics. • **PsycINFO** — Consideration of psychological and behavioural dimensions of offending. • **Web of Science** — Broad peer reviewed coverage across policing and criminology topics. • **Criminal Justice Abstracts** — Dedicated criminological literature inclusive of coverage of sex-offending typologies. • **ProQuest** — Includes the Criminal Justice Database focussed on crime-related research, law enforcement practices, and police administration. • **Google Scholar** — Grey literature, NGO reports, and regulatory publications alongside mainstream citation trails.
Track 2 — Computer Science & NLP		
Which NLP methods can be most effectively realised within policing infrastructure, and what does the evidence base reveal about model performance, explainability, and operational deployment constraints?	• IEEE Xplore • ACM Digital Library • ACL Anthology • arXiv (cs.CL, cs.AI, cs.CY) • Papers With Code • Scopus	• **IEEE Xplore** — Institute of Electrical and Electronics Engineers (IEEE) curated database for applied NLP architecture, development and deployment. • **ACM Digital Library** — Curated by the Association for Computing Machinery (ACM); similar in stature for NLP content as **IEEE Xplore**. • **ACL Anthology** — Curated by the Association for Computational Linguistics (ACL); linguistics-grounded NLP resource with a focus on language understanding and generation, essential for conversational analysis. • **arXiv** (cs.CL) — Pre-print access to novel language model research. • **Papers With Code** — Community driven repository linking code implementations to research papers, crucial for identifying reproducible NLP methods and benchmarks. • **Scopus** — Interdisciplinary bridge between technical and applied criminological literature.
Track 3 — Organisational Structures & Policing		

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Table 2.1 continued

Analytical Question	Databases	Database Justification
What structural, cultural, and institutional factors act as barriers to or enablers of NLP adoption in policing organisations, and how does change management theory account for technology resistance in law enforcement?	• Web of Science • Criminal Justice Abstracts • PsycINFO • ProQuest • Dimensions • Google Scholar	• **Web of Science** — Peer-reviewed journals across the social sciences, inclusive of law enforcement and organisational change. • **Criminal Justice Abstracts** — Focused research on policing and policing culture. • **PsycINFO** — Human and organisational literature featuring law enforcement contexts. • **ProQuest** — Includes the Criminal Justice Database focussed on crime-related research, law enforcement practices, and police administration. • **Dimensions** — Artificial Intelligence (AI) powered search tool for the identification of emerging research clusters in public-sector innovation. • **Google Scholar** — Established open source search engine, used here for grey literature and policy reports from law-enforcement agencies and inspectorates.
Track 4 — Ethics, Law & Governance		
What ethical and regulatory constraints govern NLP deployment in law enforcement, and how should proportionality, data protection, and rights-based frameworks shape the design and oversight of such systems?	• HeinOnline • SSRN • EUR-Lex • PhilPapers • ProQuest & Scopus • Dimensions	• **HeinOnline** — Links to the Law Journal Library containing peer reviewed articles on the ethical implications of AI bias. • **SSRN** — Social Science Research Network (SSRN) contains dedicated sections on criminal justice research focussed on AI and technology adoption. • **EUR-Lex** — Access to European Union (EU) legislative instruments, including the AI Act and GDPR, relevant to algorithmic oversight in law enforcement across the EU. • **PhilPapers** — Repository of academic literature exploring the intersection of ethics, law, and governance regarding police use of AI. • **ProQuest** & **Scopus** — Both offer broad peer reviewed coverage across policing and criminology topics with crossover in ethical and legal research. • **Dimensions** — AI powered search tool for the identification of emerging research clusters in ethical and legal dimensions of AI deployment.

Note. Databases assigned per track based on disciplinary indexing coverage. See Table 2.2 for full keyword sets per track.

Table 2.2: Search Keywords by Track

Core Terms	Extended Terms	Boolean Strings
Track 1 — Criminology		
• CSA and CSAM • Online grooming • Contact offending • Grooming language • Linguistic precursors • Grooming typology • Predatory grooming behaviour • Manipulation tactics	• Online-to-offline offending • Grooming progression • Grooming escalation • Grooming stages • Trust development • Sexual communication with a child • Contact sex offending precursor behaviours • CSAM to contact offending behaviour trajectory	("online grooming" OR "sexual grooming" OR "grooming behaviour") AND ("child sexual abuse" OR "CSA" OR "CSAM") ("grooming language" OR "grooming communication") AND ("child" OR "victim") AND ("sex offend*" OR "abuse") *Proximity (Scopus / WoS):* "grooming" NEAR/5 "contact offending"
Track 2 — Computer Science & NLP		
• Natural language processing • Text classification • Transformer models • BERT • RoBERTa • DistilBERT • Large language models • Explainability • Corpus annotation • Digital conversation analysis • Linguistic feature extraction • Conversational pattern analysis	• PAN shared dataset • Perverted Justice corpus • PJDB • Chat data augmentation NLP • F1 score / precision-recall classification NLP • Chat log analysis • Conversation-level classification • Hate speech detection • Toxic language detection • Conversational AI detection • Forensic linguistics • Proactive safeguarding technology	("BERT" OR "RoBERTa" OR "DistilBERT") AND ("text classification" OR "sequence classification") AND ("grooming" OR "predatory" OR "child safety") ("explainable AI" OR "XAI" OR "interpretable machine learning") AND ("NLP" OR "natural language processing") AND ("law enforcement" OR "policing" OR "criminal justice") ("PAN shared dataset" OR "Perverted Justice" OR "chat log" OR "online conversation") AND ("grooming detection" OR "predatory communication" OR "sexual solicitation")
Track 3 — Organisational Studies & Policing		

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Table 2.2 continued

Core Terms	Extended Terms	Boolean Strings
• Policing technology adoption & implementation • Organisational policing change • Law enforcement innovation • Institutional barriers in policing & public sector • Police culture and technology • AI in policing • Digital transformation in policing • Technology resistance • Evidence-based policing and technology	• Tacit vs explicit knowledge policing • Knowledge management policing • Communities of practice law enforcement • Cross-sector technology transfer • Public sector AI adoption • Police reform • Technology scepticism • Organisational inertia • Technical literacy leadership • National policing technology • Fragmented governance policing technology • Insider outsider role change	("technology adoption" OR "technology resistance" OR "organisational inertia") AND ("policing" OR "law enforcement" OR "police force") ("tacit knowledge" OR "explicit knowledge" OR "knowledge management") AND ("policing" OR "police officer*" OR "law enforcement") *Proximity (WoS):* "police culture" NEAR/8 ("technology resistance" OR "innovation")
Track 4 — Ethics, Law & Governance		
• Algorithmic bias • Algorithmic decision-making • AI ethics • Data protection • GDPR • Proportionality • Responsible AI • AI governance • Explainable AI • UK DPA	• Predictive policing ethics • Automated decision-making law enforcement • Data minimisation • Algorithmic accountability • Ethical AI deployment • Rights-based safeguarding • EU to UK regulatory divergence	("algorithmic bias" OR "automated decision-making" OR "algorithmic accountability") AND ("law enforcement" OR "policing" OR "criminal justice") AND ("ethics" OR "human rights" OR "fairness") ("data protection" OR "GDPR" OR "UK DPA") AND ("NLP" OR "machine learning" OR "artificial intelligence") AND ("policing" OR "criminal justice")
Cross-Track — Consolidated searches		

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Table 2.2 continued

Core Terms	Extended Terms	Boolean Strings
• Proactive safeguarding • Preventative intervention • Risk prediction • NLP in policing • AI child safeguarding • NLP grooming detection • Machine learning child protection • AI criminal justice deployment barriers		("natural language processing" OR "NLP" OR "text classification") AND ("grooming" OR "child sexual exploitation" OR "online predator") AND ("detection" OR "identification" OR "classification") ("NLP" OR "machine learning" OR "artificial intelligence") AND ("policing" OR "law enforcement") AND ("implementation" OR "adoption" OR "deployment") AND ("barrier*" OR "challenge*" OR "constraint*")

Table 2.3: Inclusion Criteria

Inclusion Criteria

IC1 Peer-reviewed articles, conference proceedings, and edited book chapters, 2018–present.

IC2 Grey literature from authoritative sources (National Police Chiefs' Council (NPCC), Home Office, Child Exploitation and Online Protection Centre (CEOP), National Institute of Standards and Technology (NIST), Information Commissioner's Office (ICO)), 2018–present.

IC3 NLP, machine learning, or computational linguistics applied to safeguarding, criminal justice, or analogous operational contexts.

IC4 Grooming, CSA, or online predatory behaviour with linguistic, behavioural, or empirical focus.

IC5 Organisational, cultural, or governance studies on technology adoption in policing or structurally comparable public-sector contexts.

IC6 English-language sources; non-English permitted where a peer-reviewed translation or substantive abstract provides sufficient evidential grounding with explicit methodological justification.

IC7 Pre-2018 foundational works where theoretical significance is established and no updated equivalent exists, subject to explicit justification in the synthesis narrative.

Table 2.4: Exclusion Criteria

Exclusion Criteria

EC1 Sources prior to 2018 unless meeting IC7; all exceptions recorded in the review log.

EC2 Non-peer-reviewed opinion pieces, blog posts, or media articles without evidential grounding, unless from authoritative professional or statutory bodies.

EC3 Studies focused solely on adult sexual offending with no transferable relevance to child safeguarding contexts.

EC4 Technical NLP literature without contextual application to safeguarding, criminal justice, or operational deployment (foundational instructional / code text not included)

EC5 Technology adoption studies from sectors lacking structural or cultural comparability to UK policing (e.g. private-sector contexts without analogous accountability frameworks).

EC6 Duplicate publications; where multiple versions exist, the most recent peer-reviewed version is retained.

EC7 Sources for which full text cannot be obtained via institutional access, inter-library loan, or open-access repositories within a reasonable retrieval window.

2.6 Results and filtering process

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Chapter 3

The Landscape of Child Sexual Exploitation and Policing's Response

3.1 Scale and trajectory of CSA

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3.2 The grooming to contact offending pathway

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3.3 Policing's reactive posture

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3.4 The case for technological intervention

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Chapter 4

NLP and AI in Safeguarding and Criminal Justice Contexts

4.1 NLP applications in public services

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4.2 Computational approaches to grooming language detection

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4.3 The data provenance problem

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4.4 What the technical literature leaves unanswered

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Chapter 5

Technology Adoption and Organisational Change in Policing

5.1 Policing as a technology consumer

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5.2 Tacit versus explicit knowledge cultures

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5.3 Theoretical frameworks for technology adoption

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5.4 Structural fragmentation as an implementation barrier

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Chapter 6

Synthesis: Where Technical Possibility Meets Organisational Reality

6.1 The convergence problem

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6.2 Inductive and deductive reasoning across the two tracks

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6.3 Towards a theoretical position

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Chapter 7

Conceptual Framework and Research Questions

7.1 Presenting the conceptual framework

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7.2 How the literature has shaped the framework

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7.3 Refined research aim and sub-questions

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Chapter 8

Conclusions

8.1 Summary of the argument

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8.2 Implications for professional practice

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8.3 Next steps

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Appendix A

Patch One - Upload and Feedback

Appendix B

Patch Two - Upload and Feedback

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